

MODULE 1

- Incentives for process control
- Process variables types
- selection types criteria
- Process degree of freedom
- Period of oscillation
- Damping
- Chara. of physical system
 - Resistance
 - Capacitance / capacitive
 - Combo. of both
- Elements of Process dynamics
- Types of process:
 - Dead time
 - single / multi capacity
 - self regulating / non self regulating
 - interacting / non interacting
 - linear / non linear
 - selection of control action for them
- Study of $\bar{L}_i$
 - Liquid processes
 - Gas processes
 - Flow processes
 - Thermal process
 - In resp respect to above concepts.

INCENTIVES FOR PROCESS CONTROL

- 1) suppressing influence of external disturbances.
 - disturb: effect that surroundings have on plant.
 - control mechanism needed to suppress this
 - in heater system, heating coil heat may be ~~too~~ radiated to ~~avoid~~ ~~avoid~~ envts.
- 2) Ensuring stability of process.
 - controlling unstable reactors
- 3) optimize performance of chemical process.
 - to ↑ profit

SELECTN CRITERIA FOR CHOOSING ~~VAR~~ PV

- output variable must be a direct measure of product quality.
have favourable static & dynamic chara.
- input
large effect on CV
rapid effect on CV
- MV must ~~a~~ directly affect CV, not indirectly
- reliable, accurate measurements must be possible
- measurement points must have minimum time delays & time constants
- ~~measured~~ measured vars. must have adequate degree of sensitivity.

PERIOD OF OSCILLATION

- time taken by sys. to complete one full cycle / oscillation

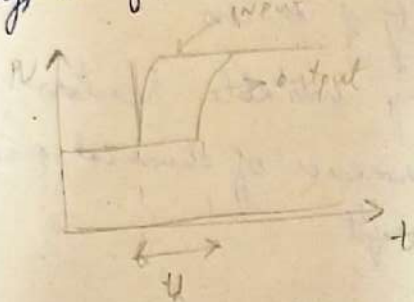
period of oscillation, $P = \frac{2\pi T}{\sqrt{1 - \xi^2}}$

ξ : damping ratio

T : characteristic time

DEAD TIME PROCESS (t_d)

- dead time is delay from when a controller output sig is issued until the measured PV ~~res~~ begins to respond.
- there's always an interval of no resp.
- caused by transportation lag, or for man, energy or info. lag or processing time



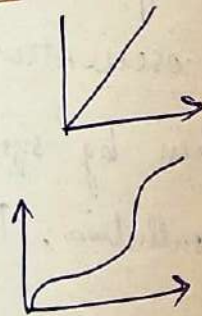
SINGLE/MULTI CAPACITY PROCESS

- * Single capacity: 1 tank, 1 time ~~capacity~~ constant.
ie 1st order TF (1st?)
 $\downarrow$
 $\begin{bmatrix} \text{SRP} \\ \text{NSR} \end{bmatrix}$
 $TF = \frac{R}{1 + ARs}, AR = \tau$
- * Multi capacity: more than 1 tank, more than 1 time constant
 $\downarrow$
interacting & non interacting ~~steps~~
depending on how single capacities are joined

LINEAR VS. NON-LINEAR SYSTEMS/PROCESSES

linear relationship b/w input & output:
eg: cylindrical tank.

non linear relationship b/w ip & op.
eg: conical tank



TF OF THERMAL SYSTEM

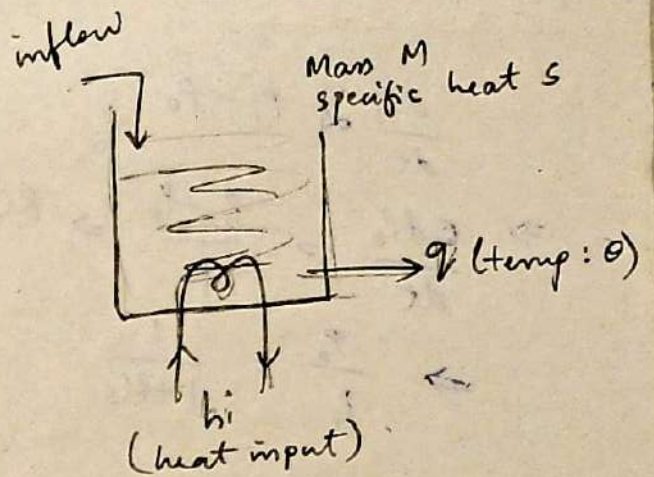
heat input rate,

$$h_i = h_1 + h_2$$

h_1 : heat stored in tank.

h_2 : heat exp. for temp of, θ

$$h_1 = MS \frac{d\theta}{dt}$$



When mass of liquid (M) is change in temp $\left(\frac{d\theta}{dt}\right)$ is $MS \frac{d\theta}{dt}$ is the heat (h_1) balance in room. specific heat (S) is the room.

$$h_2 = q_s \theta$$

When outflow q_s is temp. balance in room, heat (h_2) is the room.

Thermal resistance, $R = \frac{1}{q_s}$

$$\therefore h_2 = R\theta$$

Thermal capacitance, $C = MS$

$$h_1 = C \frac{d\theta}{dt}$$

$$\therefore h_i = C \frac{d\theta}{dt} + \frac{\theta}{R}$$

$$\Rightarrow R h_i = RC \frac{d\theta}{dt} + \theta \Rightarrow \frac{h_i}{R} = \frac{RC}{RC+1} \frac{d\theta}{dt} + \frac{\theta}{RC+1}$$

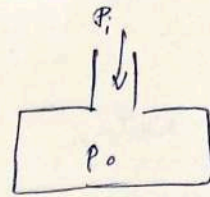
$$RC \frac{d\theta}{dt} + \theta = R h_i$$

TF OF GAS PROCESS

$$\frac{dP_o}{dt} \propto P_i - P_o$$

$$\Rightarrow \frac{C dP_o}{dt} = \frac{P_i - P_o}{R} \Rightarrow RCs P_o + P_o = P_i$$

$$\Rightarrow \frac{P_o}{P_i} = \frac{1}{1+RCs}$$



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MODULE 1

1) Self regulating vs non self-regulating process (3)

→ SRP is where PV settles at new steady state value w/o requiring external control intervention
i.e. system stabilizes itself due to physical constraints or internal feedback loop.

eg: liquid flow system w/ tank where i/f $\uparrow$ so liquid level $\uparrow$ ~~stabilizes since o/f will adjust~~
so o/f $\uparrow$, eventually level stabilizes due to gravity (it stabilizes o/f rate)

OR water heater system

Hot H_2O i/f $\uparrow$ & cold H_2O ~~down~~,
 H_2O temp $\uparrow$ ^{change} until stability
~~o/f $\uparrow$~~

→ NSRP doesn't have its PV settle at a new steady-state value on its own when i/p changes.

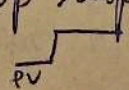
PV keeps changing

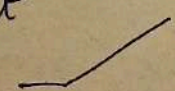
external ~~control~~ corrective action is required

eg: tank w/ no o/f

if i/f $\uparrow$, level $\uparrow$ unless corrective action is taken.

- Diff is in open loop response to step input

SRP o/p $\rightarrow$ 

NSRP o/p $\rightarrow$ 

SRP

reaches new steady state on its own after disturbance has internal feedback.

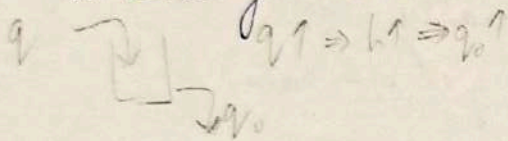
settles to finite steady value

natural eqbm exists

easier

P control enough.

inherently stable



NSRP

doesn't reach steady state on own. requires external flb.

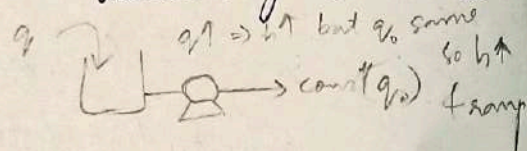
ramp-like resp. changes indefinitely

no natural eqbm.

Harder

PI / PID needed

inherently unstable.



nce of value of process degree of freedom (3)
(< 0 , > 0)

degree of freedom (DoF) refers to the no. of independent variables that must be specified in order to completely describe a system. define a process.

if independent variables are n_v
equations are n_e .

$$F, n = n_v - n_e$$

= 0, system is completely specified / exactly

red i.e. $n_v = n_e$.
ful design to prevent instability
m is controllable but no flexibility / redundancy.

manipulated vars & 3 eqns

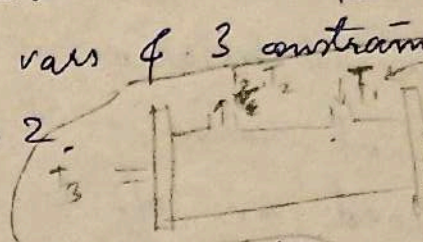
* If $n > 0$

system is under-constrained / redundant.
flexible operation possible.
performance can be optimized.
system is controllable

* no. of independently acting automatic controllers must not exceed DoF

eg: 5 manipulated vars & 3 constraints

$$DoF = 2$$



no. of vars: 6

no. of eqn = 1 (energy conservation)

$$DoF = 5$$

* If $n < 0$

system is over-specified / infeasible.

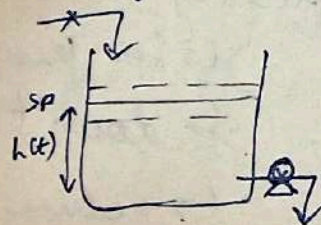
not enough vars to satisfy eqns / ~~constraints~~ constraints.

2 manipulated vars & 3 eqns

$$DoF = -1$$

11a) Explain need of automatic process control system in industry (6)

* take case of hot water level maintaining system.
we have set point (ideal water level) and an inflow and outflow w/ valves.



* if rate ~~of~~ can be controlled

Automatic control process control system is a sys. designed to automatically regulate one / more ~~processes~~ PV (temp, H_2O level, ...) w/in a desired range or at a SP w/o ~~cont~~ human intervention

- does this by continuously measuring value of PV using sensors, comparing w/ SP, control actions provided v/a actuators (valves, pumps, heaters).

9h H₂O level sys.

- if rate : manipulated var.
- level : ~~the~~ controlled variable.
- level sensor
- PI/PID controller

- SP, offset : sustained deviation

$$\text{Deviation} = \text{Actual value} - \text{SP}$$

- action taken based on offset via actuators
- PV : derived water level value.

Need.

- to ensure safety of workers.
 - ↳ in boilers / nuclear power plants, ~~not~~ ^{not} practical for workers to check conditions
- Efficiency
 - ↳ for precise regulation of PV.
 - ↳ doesn't let PV go ~~so~~ away from sp.
 - ↳ reduce energy consumption
 - ↳ minimize waste

~~see above~~ economic benefits

- integration w/ digital systems
- fast response to processes
 - ↳ pharmaceuticals.
 - ↳ food factories, etc.
 - ↳ manual response will be too slow.
- reduce workload & need for low level workers.

MOD 4 MOD 5
11 b) Classification of process variables & explanatⁿ w/ examples (8)

Quantities in a process that have to be monitored & regulated - PV.

~~Class~~

1) Controlled variable (CV)

- primary o/p of process.
- want to keep at SP.
- control sys. designed around keeping $CV = SP$.

egs

2) Manipulated variable (MV)

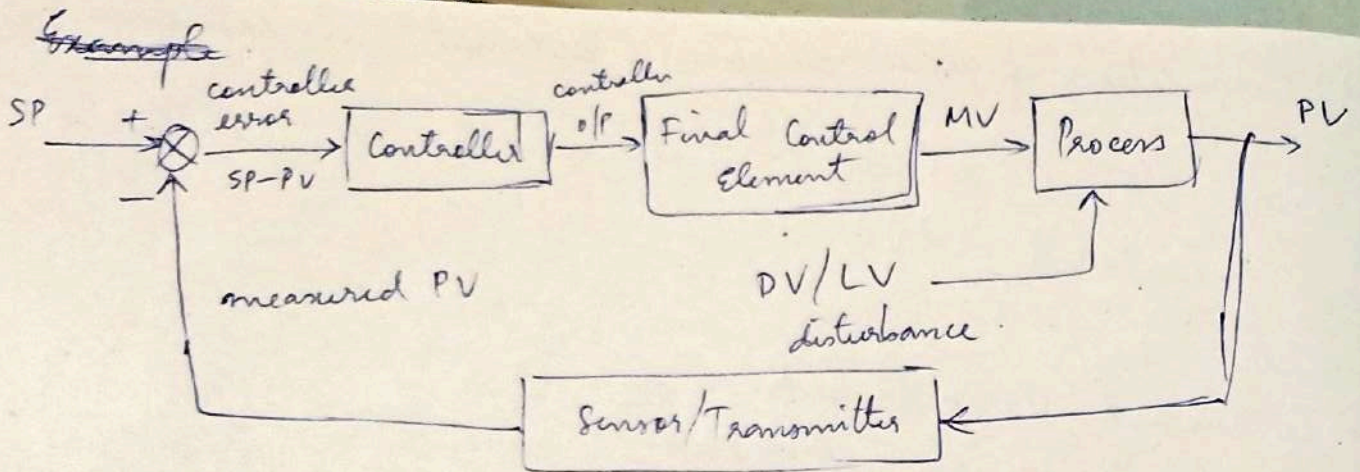
- controller adjusts to keep $CV = SP$.
- essentially process i/p.
- ~~the~~ controller modifies based on error ($SP - CV$)

3) Measured variable.

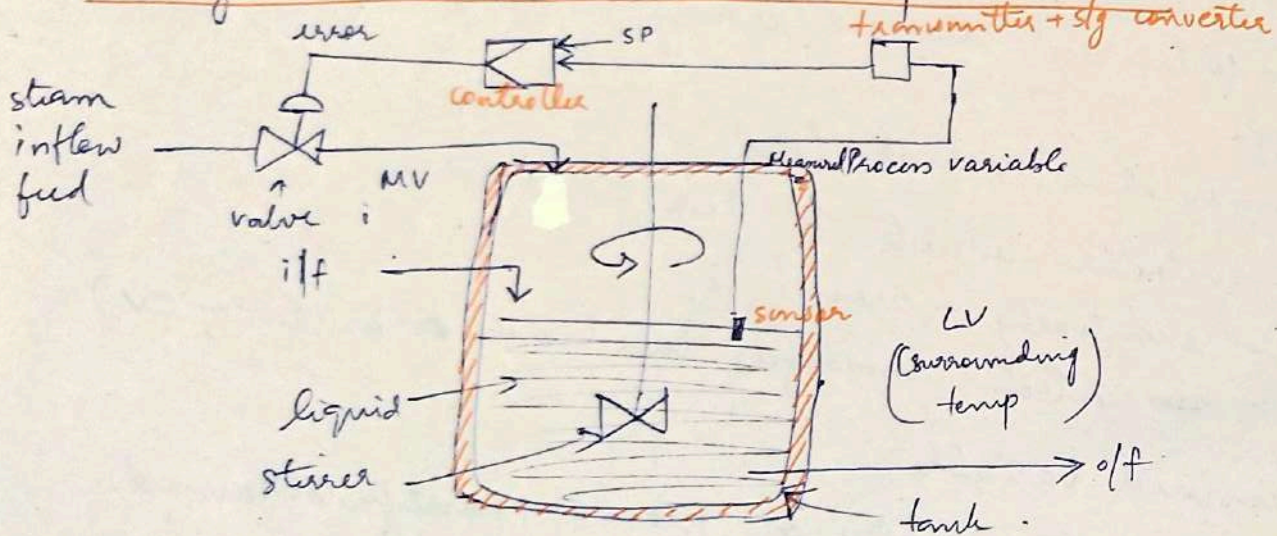
- real time value of process quantity obtained using sensors & transmitters.
- usually same as CV
- actually is sensor o/p (happens to be equal to CV some times)

4) Disturbance Variable (DV) / Load variable

- uncontrolled input
- pushes CV away from SP.
- can be measurable or not.
- must be minimized.



Controlled Stirred Tank Reactor : temp control



- cold liq. ~~also~~ conts fed into tank
- steam passed through jacket
- temp sensor inside.
- sensor $\rightarrow$ transmitter $\xrightarrow{PV}$ controller $\xrightarrow{error}$ actuator
- controls steam i/f valve.

CV : o/p fluid temp

MV : steam inflow rate

PV : temp of liquid/fluid.

PV : ambient temp changing @ liq. temp.

(2a) Process control system classifiⁿ depending on damping value.
 - explain diff types & draw typical response diagrams. (6)

2nd order system TF : $\frac{K\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$

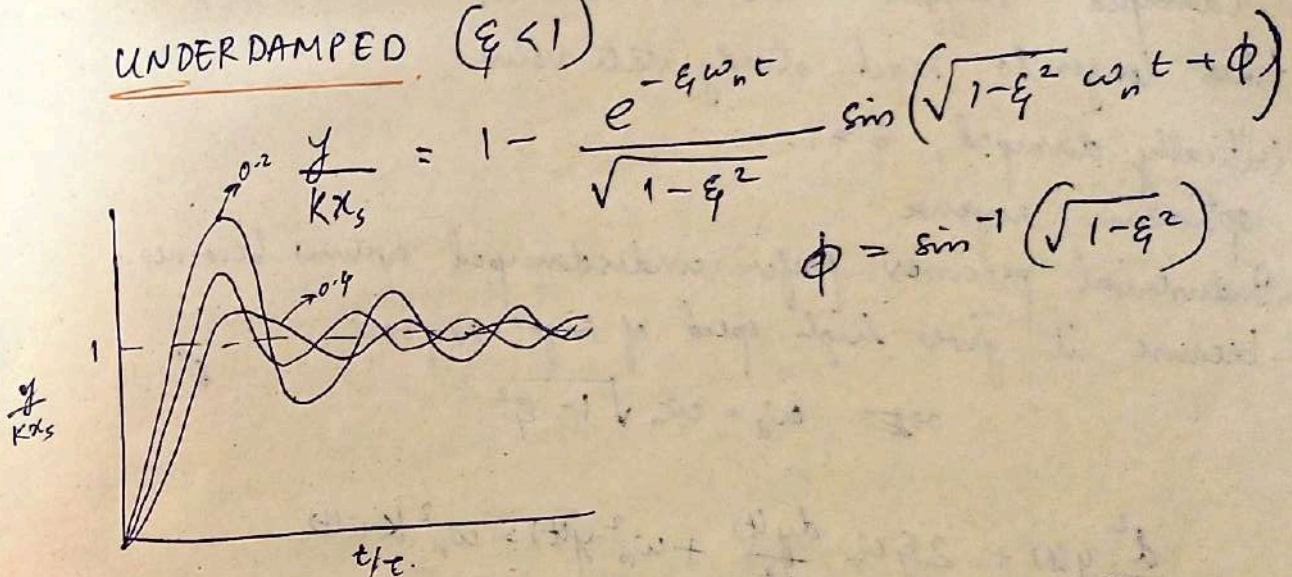
ω_n : undamped natural freq of system

ξ : damping factor

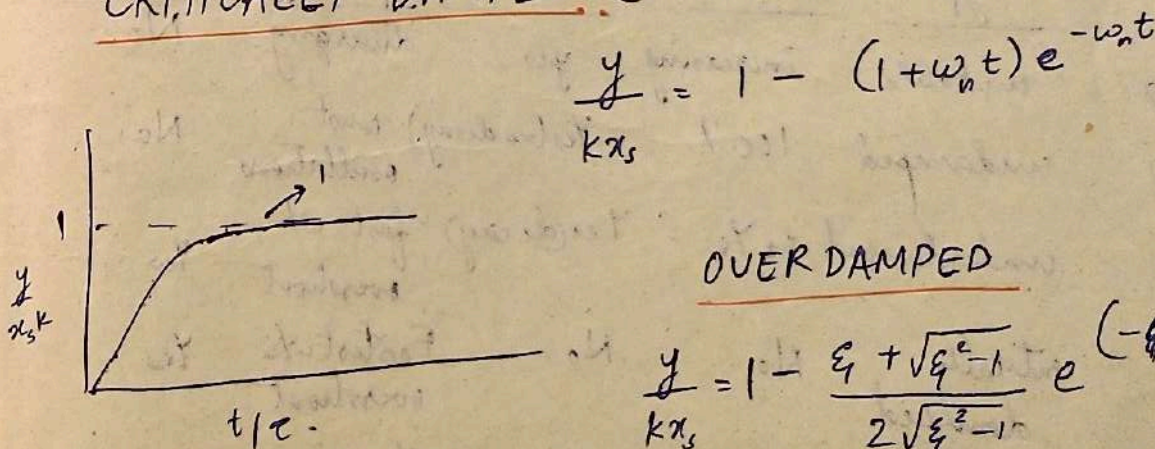
K : steady state gain

τ : time constant ($= \frac{1}{\omega_n}$)

UNDERDAMPED ($\xi < 1$)



CRITICALLY DAMPED ($\xi = 1$)



OVERDAMPED

$\frac{y}{Kx_s} = 1 - \frac{\xi + \sqrt{\xi^2 - 1}}{2\sqrt{\xi^2 - 1}} e^{(-\xi + \sqrt{\xi^2 - 1})\omega_n t}$

$+ \frac{\xi - \sqrt{\xi^2 - 1}}{2\sqrt{\xi^2 - 1}} e^{(-\xi - \sqrt{\xi^2 - 1})\omega_n t}$

- Underdamped response initially faster & reaches its ultimate value quickly but doesn't stay there. It oscillates about the steady state value. The amplitude decreases with time. The output is a damped sinusoid.

"Oscillates with progressively decreasing amplitude".

oscillatory behaviour $\uparrow$, $\zeta \downarrow$
shows overshoot

- Critically

- Overdamped - "sluggish" but no overshoot takes longer to reach steady state value

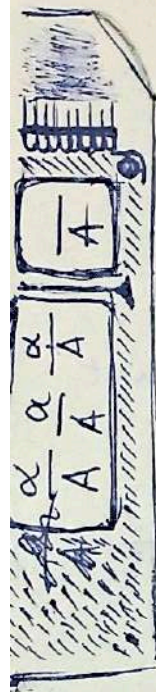
- Critically damped, $\zeta = 1$
optimum response

- Industrial processes prefer underdamped systems because it gives high speed of response.

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\frac{d^2 y(t)}{dt^2} + 2\zeta\omega_n \frac{dy(t)}{dt} + \omega_n^2 y(t) = \omega_n^2 K u(t)$$

ζ	type	overshoot	oscillations	speed	stability
$\zeta < 0$	unstable	increasing	yes	diverging	No
$\zeta = 0$	undamped	100%	Yes (no decay)	const. oscillations	No
$\zeta < 1$	underdamped	Yes	Yes (decay)	fast w/ overshoot	Yes
$\zeta = 1$	critically damped	No	No	Fastest w/o overshoot	Yes
$\zeta > 1$	overdamped	No	No	slower response	Yes



12b) Elements of process dynamics w/ necessary diagrams. (8)

1) Process Gain (K_p)

- steady state change in o/p variable per unit change in the input variable
- tells how much process change the process output will eventually for a given change in input

$$K_p = \frac{\Delta \text{output}}{\Delta \text{input}}$$

- $K_p \propto$ sensitivity
- eg: if increasing steam flow by 10% causes 5°C rise in temp,
 $K_p = 0.5^\circ\text{C}/\%$

2) Time Constant (τ)

- represents speed of process response
- time it takes for process to reach about 63.2% of total change after a sudden step input.

$$\tau \propto \frac{1}{\text{speed}}$$

- lower & smaller τ means faster spec system
- eg: a smaller heated tank w/ liquid heats quicker than a larger heater bc it has lesser liquid $\Rightarrow \downarrow \tau$

3) Dead time and Time Delay (θ)

- delay b/w Δinput & observable response in output
- during dead time, process appears unresponsive & even though the input has changed
- could be due to sensor lags, actuator lags, etc
- dead time makes control more difficult and requires predictive control strategies

eg: in a pipeline heating sys. it might take 5 mins before heated fluid reaches sensor at outlet

4) Capacitance

- ability of a process to store material, energy or ~~slow~~ charge.
- higher capacitance processes resist ~~quick~~ quick changes,
low low capacitance processes react fast
- high capacitance, larger τ , slower response.
- eg: a large tank stores a huge volume of liquid (high capacitance), while small pipe section has very little storage, (low capacitance)

5) Resistance

- opposition of the flow of material, energy or information through a system.
- higher resistance, slower flow. so slower response.
- eg: in a fluid system, a narrow pipe has high resistance to flow.

$$R = \frac{fL}{A} \Rightarrow R \propto \frac{L}{A}$$

6) Inductance (Inertia)

- tendency of a system to resist change in its motion/flow.
- systems w/ inductance (high) takes longer to start changing after an input force is applied.
- caused a lagged response beyond just capacitance effects.
- eg: a larger mechanical flywheel in a motor system takes time to speed up/slow down.

Process Element	Meaning	Effect on dynamics	Example
Process gain (K_p)	Magnitude of o/p change	High gain: more sensitive system	Temp. rise per steam flow.
Time const (τ)	Speed of response	small τ : faster response	small heater heats quickly.
Dead time (θ)	Delay before o/p reacts	increases difficulty of control	pipeline transport delay.
Capacitance	Storage ability	High capacitance: slow change	large storage tank
Resistance	opposition to flow	High resistance: slower so flow	Narrow pipe
Inductance	Inertia/resistance to change	causes sluggish start initially	Heavy flywheel.

θ : delays start of response

$R \& C$: shape how smooth or slow the rise is

L : lags initial reaction

K_p : scales mag. of o/p change

or is it sensor, actuator, controller, etc
also, good control diagram version?